

Rohit Acharya

Kathmandu, Nepal | +977-9863798002 | rohitofficial.ce@gmail.com | [LinkedIn](#) | rohitacharya.dev | [GitHub](#)

RESEARCH INTERESTS

Artificial intelligence and computational methods for construction and structural engineering: computer vision and BIM-based construction progress monitoring, digital twins, machine learning for structural health monitoring, and data-driven infrastructure assessment. Built on a foundation in computational structural mechanics, including finite element modeling of buckling and stability of advanced composite and functionally graded plates under thermal-mechanical loading, and FRP rehabilitation of reinforced concrete.

EDUCATION

National Institute of Technology (NIT) Rourkela

Odisha, India

Bachelor of Technology in Civil Engineering — CGPA: **7.96 / 10.0**

Aug 2021 – May 2025

- **Thesis:** *Buckling Performance Evaluation of Functionally Graded Graphene-Reinforced Composite Plates under Thermal-Mechanical Loads*; Supervisor: Prof. Shishir Kumar Sahu.
- **Relevant Coursework:** Finite Element Method, Advanced Structural Analysis, Mechanics of Solids, Reinforced Concrete Design, Foundation Engineering, Fluid Mechanics, Numerical Analysis, Construction Planning and Scheduling.

Kathmandu BernHardt Secondary School

Kathmandu, Nepal

Higher Secondary Certificate, Science Stream — CGPA: **3.85 / 4.0**

2018 – 2020

- Earned A+ grades in Physics, Chemistry, Mathematics, and Computer Science.

RESEARCH EXPERIENCE

Undergraduate Researcher — Computational Structural Mechanics

Jun 2024 – Dec 2024

Department of Civil Engineering, NIT Rourkela | Supervisor: Prof. Shishir Kumar Sahu

Rourkela, India

- Modeled buckling and post-buckling behavior of functionally graded graphene-reinforced composite plates under coupled thermal-mechanical loading using Classical Plate Theory and finite element analysis.
- Implemented through-thickness material gradation using power-law volume fraction distribution and Halpin-Tsai micromechanics; formulated the buckling problem as a generalized eigenvalue system, $[\mathbf{K}]\{u\} = \lambda[\mathbf{K}_G]\{u\}$.
- Conducted ABAQUS parametric simulations across 12+ variables, including gradient index, aspect ratio, width-to-length ratio, GNP volume fraction, plate thickness, number of graphene layers, temperature, and boundary conditions.
- Found that increasing GNP volume fraction from 5% to 25% improved critical buckling load by approximately 30%, while thermal exposure from 300 K to 500 K reduced capacity by approximately 15%.
- Validated mesh convergence using C3D20R second-order solid elements; studied geometric imperfection sensitivity through sinusoidal perturbation and captured post-buckling response using nonlinear static analysis.
- Benchmarked numerical trends against prior composite-plate studies, including Li et al. (2018), Feng et al. (2017), and Song et al. (2017), with close agreement.

Seminar Researcher — FRP Strengthening of Reinforced Concrete Structures

2024

Department of Civil Engineering, NIT Rourkela | CE4900: Seminar & Technical Writing

Rourkela, India

- Reviewed CFRP, GFRP, and BFRP retrofitting literature for flexural, shear, and axial strengthening of aging reinforced concrete structures.
- Compared externally bonded and near-surface mounted FRP systems; analyzed design tradeoffs for seismic and coastal durability applications.
- Derived and validated analytical formulations for flexural capacity, FRP shear contribution, and axial confinement; summarized reported flexural gains of 60–85%, shear gains up to 200%, and axial capacity gains of 50–60%.

RESEARCH OUTPUT

B.Tech. Thesis: *Buckling Performance Evaluation of Functionally Graded Graphene-Reinforced Composite Plates under Thermal-Mechanical Loads*. Department of Civil Engineering, NIT Rourkela, 2025.

Research Artifacts: ABAQUS parametric models, MATLAB/Python post-processing scripts, mesh-convergence results, and technical thesis report.

PROFESSIONAL EXPERIENCE

CloudFactory

Lalitpur, Nepal

Construction Analyst — Buildots AI Construction Progress Tracking

Jul 2025 – Present

- Evaluate AI-assisted construction progress by comparing 360° site walkthrough imagery with BIM models, schedules, and planned construction sequences.
- Perform quality audits on annotated construction datasets for hotel and large-scale residential projects; identify sequencing gaps, scope mismatches, and progress deviations.
- Contribute domain-specific review standards to improve dataset reliability for computer-vision-based construction monitoring models.

Envirocare Infrastructure Pvt. Ltd. (Shri Mahavir Ferro Alloys)	Rourkela, India
<i>Project Intern — Site Execution & Estimation</i>	<i>Jan 2025 – May 2025</i>
- Supervised civil execution works across industrial structures, including New SMS, Rolling Mill, and Pump House areas. - Prepared BOQ and Running Account bills from site measurements and structural layout drawings; coordinated with contractors, site engineers, and billing teams.	
Bureau of Indian Standards (BIS)	Guwahati, India
<i>Intern — Quality Assurance, Civil Engineering Division</i>	<i>May 2024 – Jul 2024</i>
Applied Indian Standards to material compliance testing for cement, aggregates, and steel reinforcement; assisted in document audits and compliance report analysis.	

SELECTED PROJECTS

Structural Health Monitoring Dashboard Project Website Next.js, TypeScript, Recharts	2025 – Present
- Developed a real-time SHM prototype for simulated bridge and building sensor streams, including accelerometer, strain gauge, displacement, and temperature channels. - Implemented rolling z-score anomaly detection and real-time visualization; designed the system architecture for future integration with live sensor hardware via WebSocket.	
FG-GRC Plate Buckling Simulator Project Website React, Python, FastAPI, MATLAB	2025
- Built an interactive buckling and free-vibration calculator implementing Halpin–Tsai micromechanics and Classical Plate Theory eigenvalue solution. - Computed nondimensional critical buckling load and fundamental frequency across GPL distribution patterns, including UD, FG-X, FG-O, and FG-V.	
FRP-RC Section Analyzer Project Website React, TypeScript, Tailwind	2025
- Created a dual-code design aid for FRP-reinforced concrete sections using IS 456 and ACI 440.1R concepts. - Computed cracking moment, ultimate capacity through strain compatibility, trilinear moment–curvature response, and Branson-method deflection with a transparent calculation audit trail.	

TECHNICAL SKILLS

Simulation & Structural Analysis: ABAQUS 2023, SAP2000, ETABS, STAAD.Pro, eigenvalue buckling, nonlinear static analysis, post-buckling analysis, C3D20R elements.
Programming & Scientific Computing: Python, NumPy, SciPy, Pandas, Matplotlib, MATLAB scripting, FastAPI, TypeScript, JavaScript, React, Next.js.
Civil Engineering Tools: AutoCAD, Revit/BIM concepts, BOQ and quantity estimation, construction scheduling, MS Excel.
Data & Digital Construction: Time-series analytics, anomaly detection, construction dataset QA, Buildots AI construction platform.
Documentation: Latex, technical report writing, MS Office Suite.
Languages: Nepali (Native), English (Professional/C1), Hindi (Fluent).

AWARDS & HONORS

Aspire Institute Leadership Program Scholar, 2024 — Selected for a competitive global leadership program focused on decision-making, social impact, and cross-cultural collaboration.
District Topper, Baglung, Nepal, 2018 — Ranked among the highest-performing students in the district in the Secondary Education Examination with GPA 3.95 / 4.0.
Academic Distinction, Higher Secondary Science Stream — Earned CGPA 3.85 / 4.0 with A+ grades in Physics, Chemistry, Mathematics, and Computer Science.

LEADERSHIP & SERVICE

Student Mentor	May 2023 – May 2025
<i>Institute Counselling Services, NIT Rourkela</i>	<i>Rourkela, India</i>
Provided academic, personal, and career guidance to junior students through NIT Rourkela’s structured peer-support program.	
Founder & Lead Educator	Jul 2020 – Jun 2024
<i>Learners Club, Kathmandu, Nepal</i>	
Founded a free STEM education platform during COVID-19 for school students in Nepal; produced Physics, Chemistry, and Mathematics learning content reaching 20,000+ views.	
Program Coordinator	Jun 2022 – May 2025
<i>SDG Campus Club, NIT Rourkela</i>	<i>Rourkela, India</i>
Coordinated sustainability-focused campus activities aligned with UN Sustainable Development Goals, supporting event planning, logistics, outreach, and team execution.	
Student Member	Apr 2021 – Jun 2024
<i>American Society of Civil Engineers (ASCE)</i>	

CERTIFICATIONS

Python for Data Science, AI & Development — IBM / Coursera, 2024	Foundations: Data, Data, Everywhere — Google / Coursera, 2024
Introduction to Thermodynamics — Coursera, 2023	AWS S3 Basics — AWS / Coursera, 2023